Commentary2Highlight

The same 0–10 stages, mapped to our project | 19 October 2026

Pipeline at a glance

The proposed design keeps its original stage order. The current deliverable takes commentary audio to timestamped ranked commentary. Whisper and optional translation are pretrained input components; the downstream NLP rules and PyTorch rankers are implemented by our team. Status labels below describe actual work in the project files.

0 Audio + Whisper	1 Data	2 Preprocess	3 Entities	4 Events	5 TF-IDF	6 Sequence	7 Heads	8 Inspect	9 Group	10 Export

Status key: Working = implemented and run; Partial = limited version exists; Experimental = implemented but not established as reliable; Planned = not implemented.

Stages 0–5: input, NLP, and representation

Stage 0 — Demo-only ASR (Whisper)

WORKING, WITH INPUT LIMITS — FFmpeg extracts audio; Faster-Whisper small detects language and produces timestamped speech. English output can use Whisper translate. Tested on Spanish, French, and Russian examples. First verify the video has decodable commentary audio. ASR timestamps and player spellings can be wrong.

Stage 1 — Data collection and split

PARTIAL — Timestamped SoccerNet-Echoes transcripts and SoccerNet event annotations support weak event alignment. Training splits hold out whole matches. Current checkpoint data has 380 windows from seven matches (365 weak event/routine windows and 15 reviewed windows). The supplied 2023–24 narrative CSV has unreliable row mapping and no reliable per-line timestamps, so it is not used as training truth.

Stage 2 — Preprocessing (classical NLP)

WORKING BASELINE — Project code normalizes text and captures negation, hypothetical or retrospective wording, live-action cues, emphasis, and football phrase evidence. It preserves useful punctuation and names rather than stripping everything. Translation and ASR errors still flow into this stage.

Stage 3 — Coreference / entity linking

PARTIAL: ALIAS HEURISTIC — If a verified match roster is supplied, exact aliases can be linked. A pronoun is linked only when one unique player alias occurred in the prior 30 seconds; otherwise it stays unresolved. This is not a trained general coreference model. No trained player-name or nickname recognizer exists.

Stage 4 — Event extraction / SRL

PARTIAL: LEXICAL EVENT FRAMES — Team-written patterns propose goal, penalty/VAR, card, shot/chance, and set-piece candidates, with evidence and some outcome cues. Candidate actor/team fields may use configured aliases. Full semantic-role labeling, reliable assists, score-state extraction, and robust action/outcome understanding are not implemented.

Stage 5 — Line-level representation (NLP–DL bridge)

WORKING: PROJECT-FITTED TF-IDF — The team's code learns a vocabulary and inverse-document-frequency weights from project commentary, then represents local context with sparse unigram/bigram features. The downstream MLP is initialized and trained by the team. This is not a pretrained 768-dimensional Transformer embedding.

Stages 6–10: learning, ranking, and outputs

Stage 6 — Hierarchical sequence model

EXPERIMENTAL, NOT FULL HIERARCHY — A project-built bidirectional GRU with temporal self-attention reads ordered commentary windows within a match half. It is one sequence tier, not the proposal's two-tier segment-then-match hierarchy. Its held-out eventness macro-F1 was 0.597; the TF-IDF MLP remains the stronger primary model in the reported comparison.

Stage 7 — Multi-task output heads

PARTIAL / WEAK FAMILY LABELS — Project-trained models predict event-related vs routine; the multi-task and sequence versions also predict broad event families such as goal, shot/chance, and card. Family labels are weak and sparse. There is no trained excitement score or player-identity head.

Stage 8 — Attention and inspection

PARTIAL; NOT VALIDATED EXPLANATION — The sequence model exports temporal attention weights, and TF-IDF outputs can show feature-removal effects. These help inspect model behavior but do not prove what caused a prediction; no validated explanation heatmap is claimed.

Stage 9 — Peak detection and ranking

BASIC GROUPING IMPLEMENTED — Rules and model scores rank candidate moments; nearby candidates are merged and the strongest timestamp is retained. There is no learned continuous excitement curve and no independently validated peak detector.

Stage 10 — Structured output

WORKING — The pipeline exports a timestamped transcript and ranked commentary records in Markdown, CSV, and JSON, including evidence and score details. Timestamps originate from ASR and must be checked. Automatic video clip cutting is future work.

Models used and their roles

Component	Role	Who trained it?
Faster-Whisper small	Speech recognition, language detection, timestamped segments; can translate speech to English.	Pretrained; not fine-tuned by the team.
NLLB-200 distilled 600M (optional)	Translates recognized text to English in an alternate route.	Pretrained; not fine-tuned by the team.
Event phrase rules + TF-IDF	Proposes candidate events and creates project-fitted sparse text features.	Built/fitted by the team.
PyTorch MLP / multi-task MLP / BiGRU-attention	Ranks event-related commentary; experimental models add event-family or sequence outputs.	Trained from project data by the team.

Worked project example and next work

Crystal Palace vs Manchester City — three-minute sequence demo

The demo used a video with working Spanish commentary audio. Whisper detected Spanish and produced English text. The sequence-model run returned 81 transcript segments and two ranked candidates:

• 00:01:23–00:01:26 — Chance/shot candidate; “The high shot to the front of the Crystal Palace.”

• 00:02:50–00:02:52 — Corner/set-piece candidate; text around a corner and attempted finish. The event-family head disagreed, illustrating that this output needs more work.

These are generated prototype results, not a clean final highlight video. The reported exploratory leave-one-match-out mean F1 for the TF-IDF MLP is 0.655, versus 0.553 for a linear TF-IDF baseline; the mixed labels and small dataset limit what this comparison establishes.

Priority work

1. Finish and deduplicate manual labels; include both candidate events and routine controls.

2. Keep a full match unseen until final evaluation; expand to independently reviewed matches and languages.

3. Separate event detection from subjective highlight quality and report them as different targets.

4. Create verified team/player alias lists for selected fixtures; quantify name correction rather than claiming universal name understanding.

5. Check candidate timestamps and translations against source video/audio. Add clip extraction only after timing review.

What the current system delivers

A working prototype that turns decodable football commentary audio into timestamped transcript segments and ranked commentary moments. Whisper and optional translation are pretrained input tools. The downstream NLP rules, TF-IDF representation, and PyTorch models are implemented by the team. Broad multilingual reliability, full coreference/SRL, trained player identity, and automatic clip creation remain future work.

Stage names follow the supplied “Commentary2Highlight Full Pipeline with Worked Examples” PDF. Statuses are mapped from the project code, handoff notes, data audit, and recorded model results.